

Keerti Vijay Ananth

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EXPERIENCE

Motorola Solutions *Software Engineer Intern*

Feb 2026 – Ongoing

- Engineered a Proof of Concept for a fault-tolerant distributed infrastructure system, validating continued service operability under simulated multi-node failure scenarios.
- Designed and executed CI/CD pipeline workflows to assess system resilience and failover behavior across degraded operational conditions, enabling structured evaluation of high-availability architectures.
- Researched and benchmarked distributed system fault-tolerance strategies, documenting findings to support architectural decision-making for large-scale infrastructure deployments.

BVK Exports *AIML Data Analyst Intern*

May 2025 – July 2025

- Built Python-based data pipelines using Pandas and NumPy to clean and structure multi-source datasets, reducing processing time by **40%**.
- Applied machine learning techniques (clustering, segmentation) to uncover customer patterns, improving targeting efficiency by **30%**.
- Optimized SQL queries and automated workflows to create scalable data systems, delivering actionable insights through dashboards.

PROJECTS

LLM Cost Autopilot | [Live Demo](#)

July 2026

- Built an intelligent LLM request-routing platform that classifies prompt complexity and automatically routes requests to the lowest-cost capable model with an asynchronous quality verification pipeline.
- Developed the system using FastAPI, Scikit-Learn and Streamlit, leveraging a Random Forest classifier to power routing decisions and real-time cost analytics.
- Reduced **LLM API costs by 43%** over a single-model baseline while maintaining automated quality verification through a production-ready containerized deployment on Render.

TrialLens – Clinical Trial Intelligence Tool

Aug 2026

- Engineered a **5-step LangGraph workflow** routing natural-language queries across SQL lookups, keyword-based document retrieval, and FDA adverse-event APIs, using **Llama 3.3 70B (Groq)** for intent parsing and cited response synthesis.
- Integrated **ClinicalTrials.gov** and **openFDA FAERS APIs**, implementing full-text search fallbacks for condition matching and resolving production API access issues caused by TLS fingerprinting and WAF restrictions.
- Optimized production deployment by replacing a **Chroma + sentence-transformers** retriever with a lightweight TF-IDF-style scorer, eliminating the ML runtime and resolving **512MB OOM failures**, while deploying the Dockerized service on Render with provider-agnostic LLM support.

Advanced Diabetes Prediction System

Sep 2025

- Engineered a supervised ML pipeline using Scikit-Learn to classify diabetes risk levels from clinical and lifestyle attributes with an accuracy of **94.5%**.
- Developed modular Python scripts for feature engineering, model training, hyperparameter tuning, and performance validation.
- Integrated real-world inference through a React + Node.js interface backed by AWS SageMaker, enabling instant risk-level predictions and actionable health insights.

EDUCATION

Vellore Institute of Technology

B.Tech (CSE) Health Informatics - 8.65 CGPA

Bhopal, India

Sep 2022 – Ongoing

CO-CURRICULAR

- Presented the paper “Analysis of Nanomaterial-Based Biosensor Health Monitoring Systems with Artificial Intelligence for Smart Healthcare” at ICINVENTS-25, SKCET.

TECHNICAL SKILLS

AI/Machine Learning: LangChain, LangGraph, RAG pipelines, Pytorch, Tensorflow, Computer Vision, LLM, NLP

Languages: Python, Java, Go, HTML/CSS

Frameworks: Pandas, Numpy, Scikit-Learn, Temporal, NATS, RabbitMQ

Backend/Tools: Power BI, Tableau, Linux, Git, Github, MySQL, FastAPI, REST APIs, Docker

ACHIEVEMENTS

- Winner - Ideathon, VIT Bhopal (2022).
- PR Lead & Outreach – Health-O-Tech Club (2025).
- Assisted in Python Tamil E-Content Course (2023).
- Student Coordinator – Tamil Club, VIT Bhopal (2023).